Clinical Risk Modeling and Explainability:

Mortality Prediction in Heart Failure Patients

M2 Data Science Project Report · Tabular Clinical Feature Engineering, CatBoost and 5-Fold Cross-Validation · October 2026

Abstract

This follow-on technical report describes the algorithmic and feature engineering enhancements developed for the clinical mortality risk stratification system. Addressing the variance limitations identified in the initial single-split baseline (ROC-AUC 0.759, PR-AUC 0.542), this phase introduces domain-driven interaction features capturing cardiorenal pathophysiology, vascular vulnerability, and fluid balance. Evaluated under a repeated 5-fold stratified cross-validation protocol on the leak-free cohort (299 patients, 14 predictors), model performance improved across all architectures. CatBoost achieved the highest discrimination, reaching a cross-validated ROC-AUC of 0.798 (± 0.060), a PR-AUC of 0.676 (± 0.118), and an improved Brier score of 0.178. A soft-voting ensemble combining CatBoost, Random Forest, and LightGBM demonstrated comparable stability (ROC-AUC 0.797 ± 0.071, PR-AUC 0.668). Tree SHAP attribution on the enhanced feature space revealed that the engineered `renal_cardiac_ratio` immediately became the primary driver of predicted mortality risk, aligning with clinical understanding of cardiorenal syndrome. Finally, model serialization and a production-ready FastAPI microservice were implemented.

1. Introduction

In acute and chronic heart failure, isolated laboratory measurements provide only a partial picture of patient decompensation. Myocardial dysfunction and renal impairment act synergistically in cardiorenal syndromes, where worsening heart failure impairs renal perfusion, while declining renal filtration exacerbates fluid retention and cardiac preload. The baseline phase revealed that while gradient boosting improved over regularized logistic regression, single-split evaluation suffered from substantial partition uncertainty and suboptimal precision on minority death cases (PR-AUC 0.542).

To bridge the gap toward the project specification requirements (ROC-AUC ≥ 0.85, Brier score ≤ 0.15, and clinical explainability), this iteration executes Phase 02B: (i) clinical feature engineering of compound risk ratios, (ii) 5-fold stratified cross-validation across multiple tree architectures, (iii) ensemble probability blending, (iv) post-hoc SHAP verification, and (v) serialization into a RESTful inference endpoint.

2. Feature Engineering & Clinical Rationale

Clinical Risk Modeling and Explainability: Mortality Prediction in Heart Failure Patients

2.1 Renal-Cardiac Ratio (`renal_cardiac_ratio` = `serum_creatinine / ejection_fraction`)

Clinical Rationale: Captures cardiorenal syndrome, where failing myocardial output reduces renal perfusion, while impaired renal clearance causes fluid retention and toxic metabolite accumulation that exacerbate cardiac failure. In isolation, a tree model treats high creatinine and low ejection fraction as orthogonal splits; this ratio couples them into a single physiological marker.

Statistical & Model Impact: Surpassed all raw biometrics to achieve the highest linear correlation with `DEATH_EVENT` in the dataset (**r = +0.3952** vs. +0.2943 for raw creatinine). Non-survivors presented nearly double the mean ratio (0.0596 vs. 0.0322). In SHAP analysis, it immediately became the **#1 global risk predictor**, with extreme values contributing up to +1.0 log-odds toward mortality.

2.2 Creatinine-Sodium Ratio (`creatinine_sodium_ratio` = `serum_creatinine / serum_sodium`)

Clinical Rationale: Hyponatremia (< 135 mEq/L) reflects hyperactivation of the renin-angiotensin-aldosterone system (RAAS) and non-osmotic vasopressin release, indicating circulatory volume collapse. Combining sodium with creatinine creates a sensitive index of acute renal hypoperfusion and electrolyte deregulation.

Statistical & Model Impact: Attained a correlation of **r = +0.2993** (surpassing raw creatinine and sodium at -0.1952), with non-survivors averaging 0.0136 versus 0.0087 in